

Mock Exam
Exam Preparation Exercises
Solution Sheet

Mathematics for Social Scientists

Fall 2025

Name:

Instructions

- You have 120 minutes for the exam.
- You are allowed to use one non-programmable calculator.
- **Please write your name on every extra sheet of paper you use.**
- Please write your answers clearly and indicate which question you are providing answers for on the answer sheet.
- Some questions are harder than others. The difficulty of the question **does not depend** on the question order or the number of points awarded. If a question is taking too much time, skip it and return to it after attempting the others.

Sections	1	2	3	4	Σ
Points					
/	-	-	-	-	-

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Good luck!

1 True/False and Multiple Choice Questions

For this section, on the answer sheets write down "True" or "False" for the True/False questions. Write down the answer you choose for the multiple choice questions.

True/False Questions

Problem 1. Let $A \subseteq B \subseteq C$ with $A \neq \emptyset$. Then $A \setminus C = \emptyset$.

- ☐ True
- ☐ False

Problem 2. Let $f(x) = x^2$ and $g(x) = \sin(x)$. Then $f \circ g(x) = \sin(x^2)$.

- ☐ True
- ☐ False

Problem 3. For a square matrix A , if A^2 is invertible, then A is invertible.

- ☐ True
- ☐ False

Problem 4. If events A, B satisfy $\mathbb{P}(A \mid B) = \mathbb{P}(A)$, then $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$.

- ☐ True
- ☐ False

Problem 5. Let $f(x) = \sqrt{|x|}$. Then f is differentiable at $x = 0$.

- ☐ True
- ☐ False

Problem 6. The vectors $\begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}, \begin{pmatrix} 2 \\ 4 \\ -2 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$ in $\mathbb{R}^3$ are linearly independent.

- ☐ True
- ☐ False

Problem 7. If A is $n \times n$ and $\det(A) = 0$, then the homogeneous system $Ax = 0$ has only the trivial solution ($x = \mathbf{0}$).

- ☐ True
- ☐ False

Problem 8. Consider $(\mathbb{Z}, +)$. The inverse of $n \in \mathbb{Z}$ is $(-n) \in \mathbb{Z} \quad \forall n \in \mathbb{Z}$

- ☐ True
- ☐ False

Multiple Choice Questions (2 points each)

Problem 9. You are dealt 5 cards from a deck of 52. Which of the following is the probability that you get exactly 3 kings? (*Hint: There are 4 kings in a deck of 52.*)

- ☐ $\frac{\binom{49}{2}\binom{4}{3}}{\binom{52}{5}}$
- ☒ $\frac{\binom{48}{2}\binom{4}{3}}{\binom{52}{5}}$
- ☐ $\frac{\binom{52}{4}\binom{4}{3}}{\binom{52}{5}}$

Problem 10. The determinant of the matrix $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ is

- ☐ 1
- ☐ -1
- ☐ 2
- ☒ -2

Problem 11. Which of the following is true?

- ☐ $A \cup B = A \cup C \implies B = C$
- ☐ $A \cap B = A \cap C \implies B = C$
- ☒ $B \cap C = A \implies A \subseteq C$
- ☐ $(A \subseteq B) \wedge (A \subseteq C) \implies B \cap C = A$

Problem 12. Which of the following propositional formulas is a tautology?

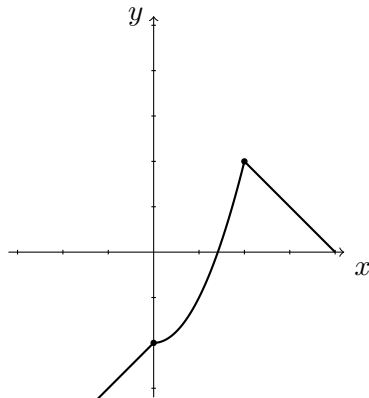
- ☒ $(P \wedge Q) \Rightarrow (P \vee Q)$
- ☐ $\neg(P \vee Q) \Leftrightarrow (\neg P \vee \neg Q)$
- ☐ $((P \Rightarrow Q) \vee (Q \Rightarrow R)) \Rightarrow (P \Rightarrow R)$
- ☐ $(P \Leftrightarrow Q) \Rightarrow (P \wedge Q)$

2 Analysis and Calculus

Problem 13. Consider the following function $f : \mathbb{R} \rightarrow \mathbb{R}$:

$$f(x) = \begin{cases} x - 2 & \text{for } x \leq 0 \\ x^2 - 2 & \text{for } 0 < x \leq 2 \\ -x + 4 & \text{for } x > 2 \end{cases}$$

- (i) Draw the function on the coordinate axis.
- (ii) Is f continuous at $x = 0$? Is it continuous at $x = 2$?



Solution.

Continuous at $x = 0$ and at $x = 2$.

Problem 14. Evaluate the following derivatives (Remember that $\frac{d^n}{dx^n}$ is the n^{th} derivative):

- (i) $\frac{d}{dx}(x^3 + 3x^2 + 3x + 1)$
- (ii) $\frac{d}{dx}(e^x + 1)$
- (iii) $\frac{d}{dy}(xy - x^2)$
- (iv) $\frac{d^2}{dx^2}\left(\frac{1}{6}x^4 + \frac{1}{2}x^2 + 1\right)$
- (v) $\frac{d}{dx}(x^2 + 5)^2$
- (vi) $\frac{d^4}{dx^4}(\sin(x))$

Solution.

- (i) $3x^2 + 6x + 3$
- (ii) e^x
- (iii) x
- (iv) $2x^2 + 1$

(v) $4x(x^2 + 5)$

(vi) $\sin(x)$

Problem 15. Evaluate the following limit **with** the L'Hospital's Rule. Explain why you can use the L'Hospital's Rule.

$$\lim_{x \rightarrow \infty} \frac{2x + 5}{1 + e^x}$$

Solution. 0 .

Problem 16. Consider the following function:

$$f : \mathbb{R}^+ \rightarrow \mathbb{R}, \quad x \mapsto x^2 \cdot \log(x)$$

- (i) Find the critical points (local max/min) of f .
- (ii) Test whether these points are minima or maxima by finding $f''(x)$. Is the function convex or concave in a neighborhood of that point??

Solution. Critical point at $x = e^{-1/2}$; local minimum; convex in a neighborhood of that point.

3 Linear Algebra

Problem 17. Let $A = \begin{pmatrix} -5 & 3 & -2 \\ 1 & 0 & -1 \end{pmatrix}$. If possible, compute the following matrix multiplications. If it is not possible, state why:

(i) $A \times \begin{pmatrix} 1 \\ 3 \\ 5 \end{pmatrix}$

(ii) $A \times \begin{pmatrix} 4 & 3 \\ 2 & 1 \end{pmatrix}$

(iii) $\begin{pmatrix} 1 & 0 \\ 0 & 0 \\ 0 & -1 \end{pmatrix} \times A$

(iv) $\begin{pmatrix} -1 & 1 \\ -1 & -1 \end{pmatrix} \times A$

Solution.

(i) $\begin{pmatrix} -6 \\ -4 \end{pmatrix}$

(ii) Not possible (dimensions mismatch)

(iii) $\begin{pmatrix} -5 & 3 & -2 \\ 0 & 0 & 0 \\ -1 & 0 & 1 \end{pmatrix}$

(iv) $\begin{pmatrix} 6 & -3 & 1 \\ 4 & -3 & 3 \end{pmatrix}$

Problem 18. Consider the following system of linear equations (SLE):

$$\frac{1}{3}x_1 + \frac{2}{3}x_2 = 1$$

$$\frac{2}{3}x_1 + \frac{2}{3}x_2 + \frac{1}{3}x_3 = 5$$

$$x_1 + \frac{7}{3}x_2 + \frac{1}{3}x_3 = 3$$

(i) Write the Gauss tableau associated with the SLE.

- (ii) We have seen in class that an SLE can be represented as $A.x = b$, where A is the coefficient matrix, $x = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$ is the vector of variables and $b = \begin{pmatrix} 1 \\ 5 \\ 3 \end{pmatrix}$ is the vector of right-hand side numbers.

Use this alternative representation and the fact that $A^{-1} = \begin{pmatrix} 5 & 2 & -2 \\ -1 & -1 & 1 \\ -8 & 1 & 2 \end{pmatrix}$ to calculate the unknown variables x_1, x_2, x_3 .

Solution.

$$(i) \left[\begin{array}{ccc|c} \frac{1}{3} & \frac{2}{3} & 0 & 1 \\ \frac{2}{3} & \frac{2}{3} & \frac{1}{3} & 5 \\ \frac{1}{3} & \frac{7}{3} & \frac{1}{3} & 3 \end{array} \right]$$

$$(ii) \ x = \begin{pmatrix} 9 \\ -3 \\ 3 \end{pmatrix}$$

Problem 19. Find the eigenvalues and eigenvectors of the matrix

$$\begin{pmatrix} 7 & 3 \\ 3 & -1 \end{pmatrix}$$

Solution. $\lambda_1 = 8$ with $v_1 = (3, 1)^\top$; $\lambda_2 = -2$ with $v_2 = (1, -3)^\top$.

Problem 20. A researcher studies how citizens' reported *political interest* (y) relates to *education* (years) and *age* (in decades). She fits the linear model

$$y = X\beta + \varepsilon,$$

where the design matrix X collects a column of ones (intercept), education in years, and age in decades, and y is a measure of political interest. The data for four respondents are:

$$X = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 3 \\ 1 & 3 & 2 \end{pmatrix}, \quad y = \begin{pmatrix} 1 \\ 3 \\ 0 \\ 3 \end{pmatrix}.$$

The OLS estimator is

$$\hat{\beta} = (X^\top X)^{-1} X^\top y$$

- (i) State the dimensions of X , y , $X^\top X$, and $X^\top y$.
- (ii) Compute $X^\top X$ and $X^\top y$.
- (iii) Compute $(X^\top X)^{-1}$.
- (iv) Compute $\hat{\beta} = (\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2)^\top$.
- (v) Compute the fitted values $\hat{y} = X\hat{\beta}$.

Solution.

- (i) $X : 4 \times 3$, $y : 4 \times 1$, $X^\top X : 3 \times 3$, $X^\top y : 3 \times 1$.

$$(ii) \quad X^\top X = \begin{pmatrix} 4 & 6 & 7 \\ 6 & 14 & 11 \\ 7 & 11 & 15 \end{pmatrix}, \quad X^\top y = \begin{pmatrix} 7 \\ 15 \\ 10 \end{pmatrix}.$$

$$(iii) \quad (X^\top X)^{-1} = \begin{pmatrix} 89/54 & -13/54 & -16/27 \\ -13/54 & 11/54 & -1/27 \\ -16/27 & -1/27 & 10/27 \end{pmatrix}.$$

$$(iv) \quad \hat{\beta} = \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix}.$$

$$(v) \quad \hat{y} = \begin{pmatrix} 1 \\ 3 \\ 0 \\ 3 \end{pmatrix}.$$

4 Probability

Problem 21. Only 1 in 100 meteors that come close to Earth actually hits the planet. To prepare for this unlikely event, physicists have developed a test that correctly classifies 80% of meteors that will hit Earth (sensitivity) and 90% of meteors that will not (specificity).

A friend at the European Space Agency tells you that there is a new meteor for which the test returned a positive result (“will hit”). He believes this means the meteor will hit the Earth with more than 50% probability. Do you agree? Justify your answer (show your work).

$$\text{Hint: } P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) = P(B \mid A)P(A) + P(B \mid A^c)P(A^c).$$

$$\text{Solution. } P(A \mid B) = \frac{0.8 \cdot 0.01}{0.8 \cdot 0.01 + 0.1 \cdot 0.99} \approx 0.0748 \text{ (about 7.5\%); Answer: No.}$$